# Filippo Lollato

contact@lollato.dev • Trento, Italy • GitHub: xupremix • Portfolio

## Profile

Computer Science graduate and Artificial Intelligence Systems MSc student with hands-on Rust experience creating Incin, an open-source machine-learning framework. I focus on typed APIs, compile-time validation, backend abstractions, readable diagnostics, and correctness checks. Seeking part-time Rust, backend, or infrastructure work alongside my studies.

## Rust Systems Project

**Incin — Rust Machine-Learning Framework** *Independent project / BSc thesis*
github.com/xupremix/incin

- Designed a typed tensor API that represents shape, data type, device, and gradient state in Rust's type system, catching invalid combinations at compile time.
- Structured the project as a multi-crate Cargo workspace, separating core tensor APIs, backend executors, procedural macros, diagnostics, data utilities, and developer tools.
- Built a CPU execution path and developer tools including `cargo incin` and `incin-lsp`; diagnostics translate complex type errors into readable tensor shapes.
- Validate operations with generated conformance vectors and differentiable operations with finite-difference gradient checks; use typed errors for unsupported runtime capabilities.

## Experience

**Teaching Assistant / Academic Tutor** *2024–25 to Present*
University of Trento Trento, Italy

- Tutor undergraduate students in Rust, Advanced Programming, and Compilers; started during the third year of my BSc.
- Explain programming concepts, help students reason through compiler feedback, and support debugging and implementation work.

## Additional Project

**ASA Autobots — Multi-Agent Delivery System** *Node.js, BDI, PDDL, LLM*

- Co-led a two-person team; contributed to system architecture and integration of BDI agents, an LLM coordinator, a PDDL planner, and service interfaces.

## Education

**M.Sc. in Artificial Intelligence Systems** *2025–26 to Present*
University of Trento Trento, Italy

**B.Sc. in Computer Science** *2022–23 to 2024–25*
University of Trento Trento, Italy
Thesis project: Incin (originally titled Kindle), a Rust machine-learning framework.

## Technical Skills

**Rust and Systems:** Rust, Cargo workspaces, type-level programming, C++20, Linux, CMake
**Architecture and Reliability:** typed APIs, backend abstractions, compile-time validation, structured errors, conformance testing, gradient checks
**Backend and Infrastructure:** REST APIs, Axum, Node.js, Docker, Git, CI/CD
**Languages:** Italian (native), English (Cambridge English B2 First, Grade A; CEFR C1)